

Business Calculus (MATH 2350)

Summer 2018

Monday, 06/25/2018

Quiz 8: Section 13.1, 13.2

Name (Print): _____

Score: _____ / 100

Section 13.1

1.

10 points

 Find $\int \frac{2}{x} dx$

2.

10 points

 Find $\int 4t^3 + 3t + 3 dt$

3. 10 points If the marginal profit of producing x surfboards is given by $P'(x) = 40 - x$, and $P(10) = 20$ where $P(x)$ is the profit in dollars, find $P(x)$.

4. 10 points Find $\int e^x + 2x^4 + \frac{2}{x} dx$.

5. 15 points (bonus) If the marginal profit of producing x items is given by $P'(x) = 40 - x$, and $P(x_0) = n$ (where $P(x)$ is the profit in dollars of producing x units, x_0 is some fixed number of units, and n is some fixed number of dollars of profit), find a general formula for $P(x)$.

(Hint: Look at problem 3 and change your values accordingly)

Section 13.2

1. In each problem, find $\int f(x) dx$.

(a) 10 points $f(x) = 24x(3x^2 + 2)^3$

(Hint: If du is too large or small, consider scaling du)

(b) 10 points $f(x) = \frac{x^2}{x^3 + 1}$

- (c) 10 points $f(x) = (x - 1)(x + 1)^5$
(Hint: If $u = f(x)$, then $f(x) - 2 = u - 2$)

- (d) 10 points $f(x) = e^{2x} + 2x + 2$

2. 10 points Find $\int 2xe^{x^2} dx$.

3. 10 points If the marginal cost for producing x of a certain item is given by $C'(x) = 4(x+10)^3$, and the fixed cost is $C(0) = 2000$ (in dollars), find the cost function $C(x)$.

4. 20 points (bonus) Find $\int \frac{20x^3 + 2x + 1}{5x^4 + x^2 + x} \ln(10x^4 + 2x^2 + 2x) dx$.